Experiment-10

Aim : To study and implement **JOIN operations** and **subqueries** using *Departments* and *Employees* tables.

Create Database

```
CREATE DATABASE Lab_Assignment_10;
▷Run
USE Lab_Assignment_10;
```

Table Structure

```
CREATE TABLE Departments (
    dept_no INT,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
▷Run | Select
CREATE TABLE Employees (
    emp_id INT,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT
);
```

Insert Sample Data

```
INSERT INTO Departments VALUES
(101, 'HR', 'Delhi'),
(102, 'Finance', 'Gurugram'),
(103, 'IT', 'Bangalore');
▷Run | Select
INSERT INTO Employees VALUES
(1, 'Jony Gautam', 50000, 101),
(2, 'Varun Chauhan', 60000, 102),
(3, 'Harsh', 55000, 103),
(4, 'Priya', 52000, 101),
(5, 'Ankit', 58000, 104);
```

(a) Display employee details where department matches in both tables (INNER JOIN)

```
SELECT *
FROM Employees E
INNER JOIN Departments D
ON E.dept_no = D.dept_no;
```

Output:

emp_id int	emp_nam varchar	emp_salary decimal	dept_no int	dept_no int	dept_name varchar	dept_location varchar
1	Jony Gautam	50000.00	101	101	HR	Delhi
2	Varun Chauhan	60000.00	102	102	Finance	Gurugram
3	Harsh	55000.00	103	103	IT	Bangalore
4	Priya	52000.00	101	101	HR	Delhi

Explanation

- INNER JOIN returns only matching records
- Employees with valid department numbers are shown
- Employee with dept_no = 104 will NOT appear

(b) LEFT OUTER JOIN

```
SELECT E.emp_name, D.dept_name
FROM Employees E
LEFT JOIN Departments D
ON E.dept_no = D.dept_no;
```

Output:

emp_name varchar	dept_name varchar
Jony Gautam	HR
Varun Chauhan	Finance
Harsh	IT
Priya	HR
Ankit	(NULL)

Explanation

- Returns **all employees**
- If department doesn't exist → shows **NULL**
- Example: Ankit will appear with NULL department

(c) RIGHT OUTER JOIN

```
SELECT E.emp_name, D.dept_name
FROM Employees E
RIGHT JOIN Departments D
ON E.dept_no = D.dept_no;  4ms
```

Output:

emp_name varchar	dept_name varchar
Priya	HR
Jony Gautam	HR
Varun Chauhan	Finance
Harsh	IT

Explanation

- Returns **all departments**
- If no employee in a department → employee name will be NULL

(d) Employees with Salary Greater than Average Salary

```
SELECT *
FROM Employees
WHERE emp_salary > (
    SELECT AVG(emp_salary) FROM Employees
);  3ms
```

Output:

emp_id int	emp_name varchar	emp_salary decimal	dept_no int
2	Varun Chauhan	60000.00	102
5	Ankit	58000.00	104

Explanation

- Subquery calculates **average salary**
- Outer query selects employees earning more than average

➢ Display Table

```
SELECT * FROM Employees;   5ms
```

emp_id int	emp_name varchar	emp_salary decimal	dept_no int
1	Jony Gautam	50000.00	101
2	Varun Chauhan	60000.00	102
3	Harsh	55000.00	103
4	Priya	52000.00	101
5	Ankit	58000.00	104

```
SELECT * FROM Departments;   6m
```

dept_no int	dept_name varchar	dept_location varchar
101	HR	Delhi
102	Finance	Gurugram
103	IT	Bangalore